

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {  
{  
    for (i = 1; i <= num;++i)  
    {  
        if (num % i== 0)  
        {  
            printf("%d ", i);  
        }  
    }  
}
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1 #include <stdio.h>
2
3 int main()
4 {
5     int n, i;
6     int counter = 1;
7
8     scanf("%d", &n);
9
10    for (i = 1; i <= n; i++)
11    {
12        counter += 2;
13
14        if (n % i == 0)
15        {
16            counter++;
17        }
18    }
19
20    printf("%d", counter);
21
22    return 0;
23 }
```

	Input	Expected	Got	
✓	12	31	31	✓
✓	25	54	54	✓
✓	4	12	12	✓

Passed all tests! ✓